

```

72.     # nextBoard is what the board will look like on the next step in
73.     # the simulation. Start with a copy of the current step's board:
74.     nextBoard = copy.copy(board)
75.
76.     # Run a single simulation step for each ant:
77.     for ant in ants:
78.         if board.get((ant['x'], ant['y']), False) == True:
79.             nextBoard[(ant['x'], ant['y'])] = False
80.             # Turn clockwise:
81.             if ant['direction'] == NORTH:
82.                 ant['direction'] = EAST
83.             elif ant['direction'] == EAST:
84.                 ant['direction'] = SOUTH
85.             elif ant['direction'] == SOUTH:
86.                 ant['direction'] = WEST
87.             elif ant['direction'] == WEST:
88.                 ant['direction'] = NORTH
89.         else:
90.             nextBoard[(ant['x'], ant['y'])] = True
91.             # Turn counter clockwise:
92.             if ant['direction'] == NORTH:
93.                 ant['direction'] = WEST
94.             elif ant['direction'] == WEST:
95.                 ant['direction'] = SOUTH
96.             elif ant['direction'] == SOUTH:
97.                 ant['direction'] = EAST
98.             elif ant['direction'] == EAST:
99.                 ant['direction'] = NORTH
100.         changedTiles.append((ant['x'], ant['y']))
101.
102.         # Move the ant forward in whatever direction it's facing:
103.         if ant['direction'] == NORTH:
104.             ant['y'] -= 1
105.         if ant['direction'] == SOUTH:
106.             ant['y'] += 1
107.         if ant['direction'] == WEST:
108.             ant['x'] -= 1
109.         if ant['direction'] == EAST:
110.             ant['x'] += 1
111.
112.         # If the ant goes past the edge of the screen,
113.         # it should wrap around to other side.
114.         ant['x'] = ant['x'] % WIDTH
115.         ant['y'] = ant['y'] % HEIGHT
116.
117.         changedTiles.append((ant['x'], ant['y']))
118.
119.     board = nextBoard
120.
121.
122. def displayBoard(board, ants, changedTiles):
123.     """Displays the board and ants on the screen. The changedTiles
124.     argument is a list of (x, y) tuples for tiles on the screen that
125.     have changed and need to be redrawn."""
126.

```